EXAMPLE OF ANALYSIS TECHNIQUES FOR SOME SPECIFIC DOMAIN

1. **E-commerce and Commerce Data Analysis**

understanding customers, sales, products, and business performance.

Main objective: Increase revenue and profit, understand customers, improve product performance, and optimize operations.

Type of analysis	What you analyze	Example question
Sales analysis	Revenue, orders, units sold, profit	Which products generate the most revenue?
Customer behavior analysis	Clicks, searches, browsing, purchases	Why do customers abandon their carts?
Customer segmentation	Demographics, behavior, spending	Which customers are most valuable?
Marketing analysis	Campaigns, ads, discounts, traffic	Which marketing channel generates the best returns?
Product analysis	Ratings, reviews, returns, sales	Which products should we promote or discontinue?
Pricing analysis	Prices, discounts, demand, margins	What happens to sales when prices change?
Inventory analysis	Stock levels, demand, lead times	Which products may run out of stock?
Customer retention analysis	Repeat purchases, churn, cohorts	Why do customers stop buying?
Delivery and logistics analysis	Shipping times, costs, failed deliveries	How can we reduce delivery delays?
Fraud analysis	Orders, payment patterns, unusual activity	Which transactions appear suspicious?

Important analytical techniques

- **RFM analysis**: Segments customers using Recency, Frequency, and Monetary value.
- Cohort analysis: Compares groups of customers acquired during different periods.
- Market basket analysis: Identifies products frequently purchased together.
- Funnel analysis: Tracks the journey from product view to purchase.
- **A/B testing**: Compares two versions of a website, offer, or checkout flow.
- Demand forecasting: Predicts future sales and inventory requirements.
- Customer lifetime value (CLV): Estimates the long-term value of a customer.
- **Churn prediction**: Estimates which customers are likely to stop purchasing.
- Price elasticity analysis: Estimates how demand changes when prices change.

2. Finance and Economics Data Analysis

Main objective: Understand financial performance, manage risk, evaluate investments, measure economic conditions, and support financial decisions.

Finance and economics are related, but they are not identical. Finance often focuses on firms, investors, banks, and financial markets. Economics focuses more broadly on consumers, businesses, markets, governments, and entire economies.

A. Finance analytics

Type of analysis	What you analyze	Example question
Financial performance	Revenue, costs, margins, cash flow	Is a company becoming more profitable?
Investment analysis	Stock prices, returns, volatility	Which assets have performed better relative to risk?
Credit risk analysis	Credit history, income, repayment data	Which borrowers are likely to default?
Fraud detection	Transactions, transfers, payment patterns	Which transactions are potentially fraudulent?
Portfolio analysis	Asset allocation, returns, correlations	How can portfolio risk be managed?
Cash flow analysis	Inflows, outflows, working capital	Will a business have enough cash?
Financial forecasting	Revenue, expenses, earnings	What might the next quarter's financial performance be?
Customer profitability	Fees, balances, service costs	Which customer segments are most profitable?
Insurance analytics	Claims, premiums, loss history	How should insurance risk be assessed?
Regulatory and compliance analysis	Transactions, customer records, risk indicators	Are there unusual patterns that require investigation?

B. Economics analytics

Type of analysis	What you analyze	Example question
Macroeconomic analysis	GDP, inflation, unemployment	Is the economy expanding or slowing?
Microeconomic analysis	Consumer and firm decisions	How do people respond to price changes?
Labor market analysis	Wages, employment, vacancies	Which factors influence wage growth?
Trade analysis	Imports, exports, exchange rates	How do exchange rates relate to trade flows?
Public policy analysis	Taxes, subsidies, public spending	Did a government policy achieve its intended effect?

Poverty and inequality analysis	Income, wealth, household surveys	How does inequality change over time?
Consumer price analysis	Prices, household expenditure	Which categories contribute most to inflation?
Regional development analysis	Income, infrastructure, employment	Which regions are growing faster?
Economic forecasting	GDP, inflation, demand, interest rates	What are the plausible economic scenarios?
Causal economic analysis	Policies, interventions, natural experiments	Did a policy cause an improvement in employment?

Important analytical techniques

- **Time-series analysis**: Studies trends, cycles, seasonality, and changes over time.
- **Regression analysis**: Examines relationships between variables, such as interest rates and investment.
- Econometrics: Applies statistical methods to economic questions and data.
- Risk modeling: Estimates default probabilities, losses, or financial exposure.
- Portfolio optimization: Evaluates combinations of investments under specified risk constraints.
- Event studies: Investigate how markets respond to events such as earnings announcements or policy changes.
- Causal inference: Estimates the effect of an intervention using methods such as difference-in-differences or regression discontinuity.
- Scenario and stress testing: Examines how outcomes could change under adverse economic or financial conditions.

3. A comparison of some domains

a quick way to understand how the same analytical technique can be used in different fields.

Analytical technique	E-commerce	Finance and economics	Computer science	Engineering
Descriptive analysis	Sales performance	Financial performance	System performance	Production performance
Exploratory data analysis	Customer behavior	Market patterns	Logs and datasets	Sensor measurements
Time-series analysis	Sales forecasting	Stock prices, inflation	Traffic and latency	Temperature, vibration
Regression analysis	Sales drivers	Economic relationships	Performance prediction	Process modeling
Classification	Customer segments	Fraud and credit risk	Spam and intrusion detection	Defect classification

Clustering	Customer gr oups	Market regimes	User or network g roups	Operating-condition groups
Anomaly detec tion	Suspicious o rders	Unusual transactions	System attacks or errors	Abnormal sensor read ings
Causal analysis	Impact of dis counts	Impact of economic p olicies	Effect of software changes	Effect of process setti ngs
Optimization	Pricing and i nventory	Portfolio allocation	Resource allocati on	Production and energ y efficiency
Predictive mod eling	Churn and de mand	Default risk and econ omic forecasts	Failure and error prediction	Maintenance and qual ity prediction

The techniques overlap significantly. What makes an analysis domain-specific is the problem, the data, the assumptions, and how the results will be used.

4. Other important domains worth exploring

Beyond these areas, data analysis is widely used in other fields.

Healthcare and medicine:
Patient outcomes, disease trends, clinical trials, hospital capacity, medical imaging, and treatment effectiveness.

Transportation and logistics
Traffic forecasting, route optimization, fleet utilization, delivery delays, and fuel consumption.

Education
Student performance, learning outcomes, course completion, dropout prediction, and teaching effectiveness.

Agriculture and environment
Crop yield prediction, soil analysis, weather impacts, water usage, climate trends, and pollution monitoring.